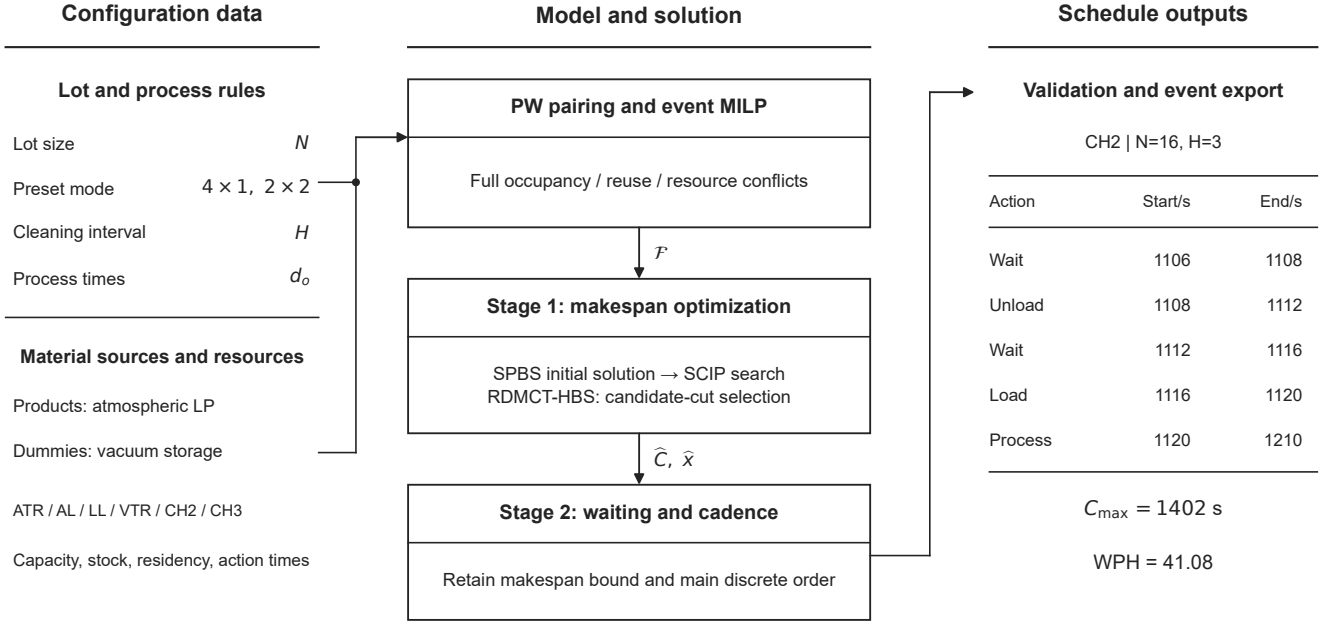

Configuration data
Lot and process rules
Lot size
N
Preset mode
4 × 1, 2 × 2
Cleaning interval
H
Process times
d_o
Material sources and resources
Products: atmospheric LP
Dummies: vacuum storage
ATR / AL / LL / VTR / CH2 / CH3
Capacity, stock, residency, action times
Model and solution
PW pairing and event MILP
Full occupancy / reuse / resource conflicts
$\mathcal{F}$
Stage 1: makespan optimization
SPBS initial solution → SCIP search
RDMCT-HBS: candidate-cut selection
$\widehat{C},\ \widehat{x}$
Stage 2: waiting and cadence
Retain makespan bound and main discrete order
Schedule outputs
Validation and event export
CH2 | N=16, H=3
Action Start/s End/s
Wait 1106 1108
Unload 1108 1112
Wait 1112 1116
Load 1116 1120
Process 1120 1210
$C_{\max} = 1402$ s
WPH = 41.08